Solution Requirements

Date	12 July 2026
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	**Requirement Category**	**Requirement Description**	**Priority (High/Medium/Low)**
1	Authentication	*Users can securely access the Flask application to view dashboards and reports.*	High
2	Authorization Levels	*Admin can manage data and dashboards, while users have read-only access to reports.*	Medium

3	External Interfaces	*Integrates Microsoft SQL Server, Tableau Public, and Flask web application.*	High
4	Transactions Processing	*Retrieve, process, and display electricity consumption data from the database.*	High
5	Reporting	*Generate interactive dashboards, stories, and analytical reports.*	High
6	Business Rules	*Validate dataset integrity, calculate yearly and regional consumption, and maintain consistent analysis.*	High
7	Compliance	*Ensure secure handling of project data and follow standard database management practices.*	Medium
8	*Other*	*Support filtering by year, state, region, and consumption trends.*	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	*Dashboards and reports should load quickly.*	*Response time < 3 seconds*
2	Scalability	*Support future datasets with additional years and states.*	*Easy dataset expansion*
3	Security & Data Privacy	*Protect database access and prevent unauthorized modifications.*	*Secure database connection*
4	Reliability & Availability	*Ensure the application and dashboards remain accessible.*	*99% availability*
5	Usability & Accessibility	*Provide an easy-to-use interface with responsive dashboards.*	*User-friendly navigation*

6	*Other*	*Maintain clean, reusable, and well-documented code.*	Easy maintenance and updates